

MicroDuck Arm v1-C Engineering Baseline

Mechanical, electrical, controls, and hardware review

2026-09-13

Release statement

ENGINEERING CONFIGURATION BASELINE - NOT FABRICATION READY

This book defines the review baseline for the MicroDuck single-arm v1-C cartridge. It is not a manufacturing drawing set, wiring release, qualified power architecture, mobile controller design, safety certification, or record of completed physical tests.

The baseline fails closed:

- fabrication release: **BLOCKED**;
- powered bench release: **BLOCKED pending staged authorization**;
- robot/mobile release: **BLOCKED**;
- physical evaluations: **NOT RUN**;
- candidate A/B qualification: **NOT RUN**.

Actual outcomes belong in the separate leader-owned `RESULTS.en.md`. A missing result, blank evidence field, simulation result, estimate, or v1-B result never becomes a v1-C pass.

1. Authority and provenance

The current configuration source is `hardware/microduck-arm-v1c/spec/design.json`. The configuration validator, environment, model adapter, and safety-state model are:

- `microduck_arm_v1c/config.py`;
- `microduck_arm_v1c/env.py`;
- `microduck_arm_v1c/model.py`;
- `microduck_arm_v1c/safety.py`.

Vendor-source retrieval and hashes are recorded in `hardware/microduck-arm-v1c/vendor/SOURCE-MANIFEST.md`. The retained Pololu D24V60F5/D24V90F5 geometry files are references only; they do not approve a regulator or its project-specific integration. ROBOTIS drawing links were located, but their advertised binary drawings were not retained because the retrieval returned landing-page HTML.

The older `docs/microduck-arm-v1/` package is v1-B reference material. v1-C inherits nominal kinematic lengths and the legacy simulation geometry where the current specification says so. It does not inherit v1-B evaluations, release status, old floor-level task success, or old power decisions.

Official candidate references:

- <https://emanual.robotis.com/docs/en/dxl/x/xl330-m288/>
- <https://emanual.robotis.com/docs/en/dxl/x/xl330-m077/>
- <https://www.pololu.com/product/2866/resources>
- <https://mujoco.readthedocs.io/en/stable/XMLreference.html>

2. v1-C configuration

2.1 Morphology and control contract

The robot model has ten leg joints plus five arm actuators. The arm has four pose joints and one gripper actuator:

Joint	Function	Action index	Provisional bus ID
J1 / M1	base yaw	10	21
J2 / M2	shoulder pitch	11	22
J3 / M3	elbow pitch	12	23
J4 / M4	wrist pitch	13	24
J5 / M5	gripper drive	14	25

IDs 21–25 are configuration placeholders only. The real IDs, physical pin mapping, firmware, actuator identity, and calibration are unverified. Real motor I/O is not implemented or approved by this baseline.

The simulation control period is 20 ms with 2 ms physics steps. The arm slew screening limit is **0.4** rad/s; the leg limit is 6 rad/s. These are simulation screening values, not qualified hardware speed or acceleration limits.

2.2 Candidate configurations

Candidate	M1	M2	M3	M4	M5	Motor mass
A, default pending qualification	M288	M288	M288	M288	M288	5 x 18 g
B, research candidate	M288	M288	M288	M077	M077	5 x 18 g

The two candidate motor types have the same documented 18 g mass:

Candidate motor	Gear ratio	Stall torque at 5 V	No-load speed at 5 V	Continuous rating
XL330-M288-T	288.4	0.52 N m	103 rpm	unknown
XL330-M077-T	77.5	0.215 N m	383 rpm	unknown

Stall torque and no-load speed are not continuous operating points. Candidate B is not automatically superior in energy, output inertia, backdrive behavior, tracking, or disturbance response. Release requires measured output inertia and backdrive evidence, matched-task energy and thermal tests, a qualified continuous torque-speed envelope, and A/B tracking and disturbance tests.

3. Geometry and changed task scene

Nominal review geometry:

Parameter	Baseline value	Release meaning
shoulder-to-elbow	55 mm	kinematic axis distance only
elbow-to-wrist	50 mm	kinematic axis distance only
wrist-to-TCP	30 mm	kinematic distance only

Parameter	Baseline value	Release meaning
summed geometric reach	135 mm	not a rated load radius or safe workspace
shoulder pivot	local [0, 0, 18] mm	v1-C change, aligned to shoulder motor envelope center
motor clearance envelope	20 x 26 x 34 mm	not a mounting drawing
jaw pivot spacing	24 mm total	gear-center candidate only
jaw length	30 mm nominal	real section and mounting TBD
modeled pad size	24 x 5 x 8 mm	physical pad properties unmeasured

The current task scene is `raised_tabletop_v1_c`:

- tabletop top: $z = 145$ mm;
- object half-height: 6 mm;
- initial object center: [115, -18, 151] mm;
- goal center: [125, 24, 151] mm;
- transfer waypoint height: $z = 190$ mm.

This is a changed scenario. The earlier 95 mm or floor-level arrangement created a low target with mount-collision risk. The raised-table scene is still provisional simulation geometry, not a released physical table, fixture, or collision-clearance result.

The rectangular pad model uses MuJoCo half-sizes [12, 2.5, 4] mm with simulation contact parameters `solref=".004 1"` and `solimp=".95 .99 .001"`. Those values do not identify a real elastomer, durometer, friction coefficient, compression curve, contact stiffness, adhesive, wear life, or grip rating.

4. Mechanical architecture and real load path

The required physical load path is:

payload -> contact pads -> rigid jaws -> jaw shafts -> jaw bearings -> gripper housing -> independent wrist shaft and bearings -> forearm -> elbow shaft and bearings -> upper link -> shoulder shaft and bearings -> base-yaw shaft and bearings -> trunk interface -> fixture or robot structure.

Motor output splines and horns transmit actuator torque into this path. They must not be treated as complete radial, axial, bending, or overturning support without explicit manufacturer evidence for the measured external loads.

4.1 Joint hardware required before release

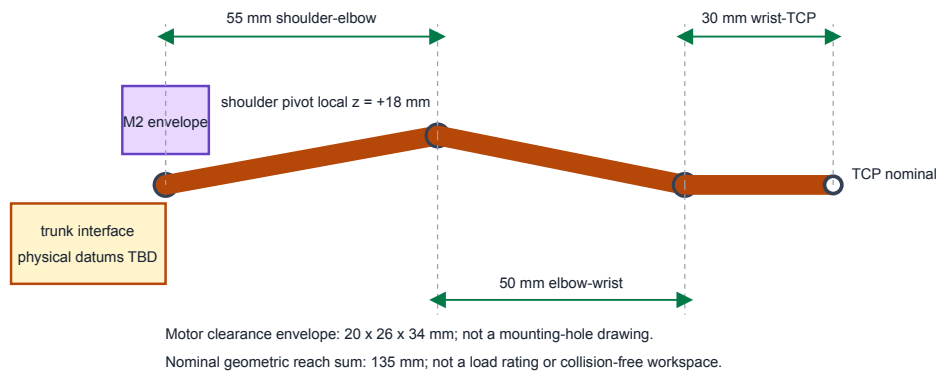
Every J1-J4 assembly requires a controlled definition for:

- structural shaft material, diameter, shoulders, runout, and fatigue;
- two-plane radial/axial reaction support or a justified equivalent;
- bearing manufacturer part number, static/dynamic ratings, fits, preload/endplay, lubrication, and contamination protection;
- exact servo horn/hub part number, spline identity, pilot, bolt circle, center screw, allowable torque, and fit;
- spacers or shims controlling axial stack without bearing side-load;
- shaft retainers, nuts, circlips, threads, locking, and pull-off evidence;
- bracket and link material, process, section, tolerances, and proof load;
- structural fastener part numbers, grades, lengths, thread engagement, tightening torque, locking, witness marking, access, and reuse rule;

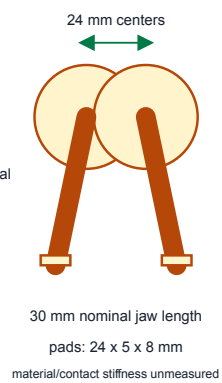
MicroDuck arm v1-C - mechanical dimension overview / 机械尺寸概览

CONCEPT - NOT SCALE-CONTROLLED - NOT FABRICATION READY / 概念图，非比例控制，不可加工

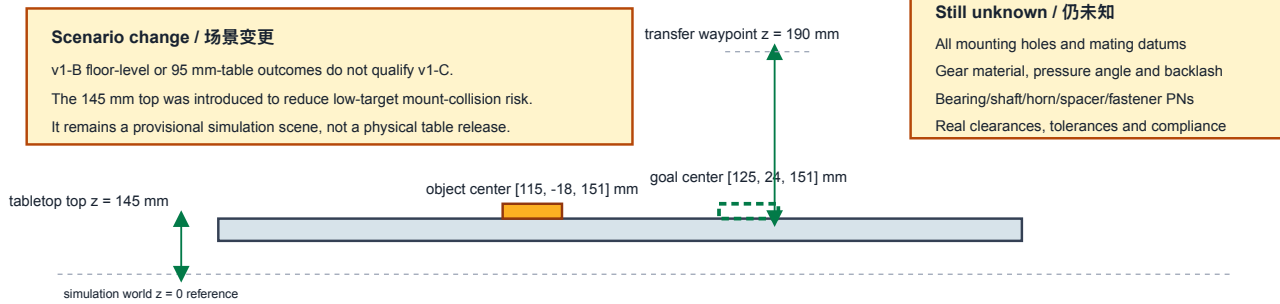
Nominal side view / 标称侧视图



Jaw plan / 夹爪俯视图



Changed v1-C raised-table scene / v1-C 已变更的高桌面场景



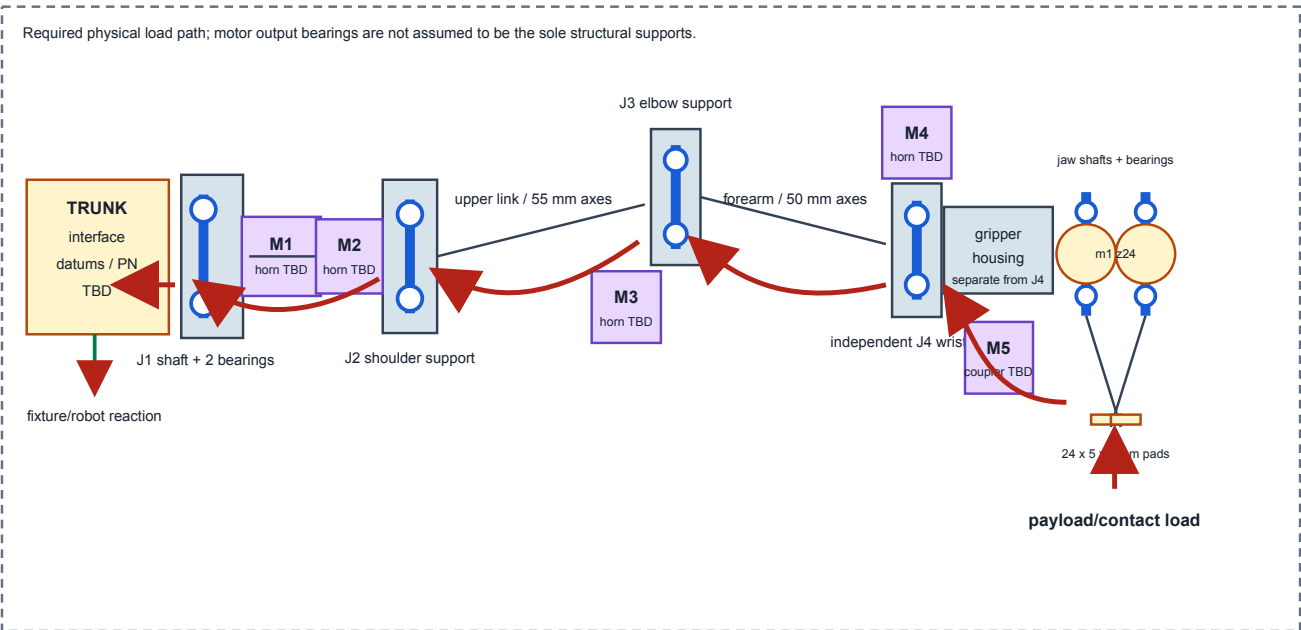
Dimensions shown are configuration values for review. Measure real hardware and issue controlled drawings before fabrication.

Figure 1: Mechanical dimension overview

MicroDuck arm v1-C - structural load path / 承力路径

CONCEPT REVIEW ONLY - NOT FABRICATION READY / 仅供概念评审，不可用于加工

Bearings, shafts, horns, spacers, retainers, fasteners, materials, fits and part numbers are TBD.



Joint stack required at J1-J4 / 每个关节所需堆叠

1. Structural shaft with shoulders and documented retention - PN TBD
2. Two separated reaction bearings or justified equivalent - PN/fits TBD
3. Servo horn/hub transfers torque; it does not define the full load path - PN TBD
4. Spacers/shims control endplay without side-loading bearings - dimensions TBD
5. Fasteners, locking, thread engagement and service limits - grades/torques TBD
6. Material, section, tolerance, fatigue and proof-load evidence - all unresolved

Release blockers / 放行阻塞项

Gear candidate: module 1 mm, 24/24 teeth,
24 mm center distance, ratio -1.

Unknown: material, pressure angle, face width,
profile shift, backlash, strength and wear.

Unknown: horn, shaft/bearing PNs, fits and retention.

Required: measured inertia and $\geq 3\times$ continuous torque.

Physical release: BLOCKED / 实体放行: 阻塞

Figure 2: Structural load path

- cable passages, bend radius, strain relief, flex life, and full-motion clearance.

All these part numbers remain TBD. The motor envelope and inherited visual mesh must not be used to infer hole positions or mating faces.

4.2 Independent wrist and jaw transmission

The wrist is its own structural joint. J4 requires an independent shaft, bearing pair, carrier, horn interface, and axial retention. The gripper housing attaches to the wrist output.

The gripper is a separate mechanism. M5 drives one jaw shaft through an explicit horn/coupler/transmission. The second jaw shaft is passive and driven by the equal external gear pair. Each jaw shaft requires its own support and retention. The design must not silently combine the wrist shaft, driven jaw shaft, and passive jaw shaft.

4.3 Gear definition

The current kinematic candidate is:

- module: 1 mm;
- driven gear: 24 teeth;
- passive gear: 24 teeth;
- nominal center distance: $m(z_1 + z_2)/2 = 1 \text{ mm}(24 + 24)/2 = 24 \text{ mm}$;
- ratio: -1;
- left driven jaw range: $[-0.32, 0] \text{ rad}$;
- passive jaw relationship: equal magnitude, opposite rotation.

This is not a gear release. Material, pressure angle, face width, profile shift, root fillet, quality grade, backlash, center-distance tolerance, lubrication, wear, allowable tooth load, hub geometry, shaft fit, retention, guarding, and debris behavior are unknown. The mechanism remains blocked until those items and physical cycle/load evidence are complete.

5. Engineering BOM and mass baseline

The controlled row-level BOM is ENGINEERING-BOM.csv. It includes ID, quantity, function, candidate, revision, mass, CoM, interface, status, required evidence, owner, and gate for motors, links, brackets, bearings, shafts, horns, spacers, retainers, fasteners, independent wrist parts, jaw transmission, gears, jaws, pads, harness, controllers, and power/safety hardware.

5.1 Corrected mass target

Category	Target or estimate
five motors	90 g
links	20 g
brackets	10 g
gripper	10 g
trunk interface	10 g
wiring and fasteners	10 g
reserve	10 g
controlled total	160 g

The user-provided category budget summed to 170 g. The controlled v1-C target is corrected to 160 g. Release requires a closed as-built mass ledger at or below 160 g; the reserve cannot become negative or conceal missing items.

The 70 g relocated-electronics, 80 g dedicated-pack, 25 g power/communication, and 8 g sensor-shell values inherited from v1-B are simulation placeholders outside the v1-C cartridge BOM. They are not purchased parts or measured packaging.

5.2 Inertia estimates

The following diagonal body-frame tensors are box estimates from the current specification. They are neither measured principal inertias nor a released assembled inertia model.

Component	Mass	Box size (m)	Estimated diagonal inertia (Ixx, Iyy, Izz) kg m ²
M1	0.018 kg	0.020 x 0.026 x 0.034	(2.748e-6, 2.334e-6, 1.614e-6)
M2	0.018 kg	0.020 x 0.026 x 0.034	(2.748e-6, 2.334e-6, 1.614e-6)
M3	0.018 kg	0.020 x 0.026 x 0.034	(2.748e-6, 2.334e-6, 1.614e-6)
M4	0.018 kg	0.020 x 0.026 x 0.034	(2.748e-6, 2.334e-6, 1.614e-6)
M5	0.018 kg	0.020 x 0.026 x 0.034	(2.748e-6, 2.334e-6, 1.614e-6)
links	0.020 kg	0.090 x 0.012 x 0.008	(3.46667e-7, 1.3606667e-5, 1.374e-5)
brackets	0.010 kg	0.050 x 0.040 x 0.012	(1.453333e-6, 2.203333e-6, 3.416667e-6)
gripper	0.010 kg	0.040 x 0.030 x 0.020	(1.083333e-6, 1.666667e-6, 2.083333e-6)
interface	0.010 kg	0.040 x 0.030 x 0.008	(8.033333e-7, 1.386667e-6, 2.083333e-6)
wiring/fasteners	0.010 kg	0.080 x 0.020 x 0.010	(4.16667e-7, 5.416667e-6, 5.666667e-6)
reserve	0.010 kg	none	no inertia estimate

The principal-inertia release gate is explicitly failed. Before S6, weigh the complete assembly, measure CoM in at least three orientations, identify each component pose, and derive or measure the assembled inertia with uncertainty.

6. Load, torque, and structural margin

For each joint:

```
tau_g,j = sum_i(m_i * g * horizontal_lever_arm_i_about_joint_j)
tau_required_continuous,j >= 3 * max_released_pose(tau_g,j)
```

The structural and actuator review must then include acceleration, reflected inertia, contact loads, cable forces, friction, backlash impact, disturbance, and thermal duty cycle. The factor of three is a minimum continuous-torque gate, not permission to divide stall torque by three.

A deliberately conservative bound places the full 160 g cartridge and 20 g payload at the full 135 mm geometric reach:

$$3 * 0.18 \text{ kg} * 9.80665 \text{ m/s}^2 * 0.135 \text{ m} = 0.715 \text{ N m}$$

That bound is not the actual shoulder requirement because real mass is distributed. It does show that neither candidate has release evidence: M288 lists 0.52 N m stall torque, M077 lists 0.215 N m stall torque, and qualified continuous ratings are unknown. The simulation force limit of 0.1 N m is also only a screening assumption.

7. Power, wiring, and controller architecture

MicroDuck arm v1-C - power and data architecture / 电源与数据架构

CONCEPT ONLY - NO QUALIFIED PACK, REGULATOR, PROTECTION OR MOBILE CONTROLLER

Default bench source is external regulated 5 V. Shared-pack and separate-pack paths are mutually exclusive studies.

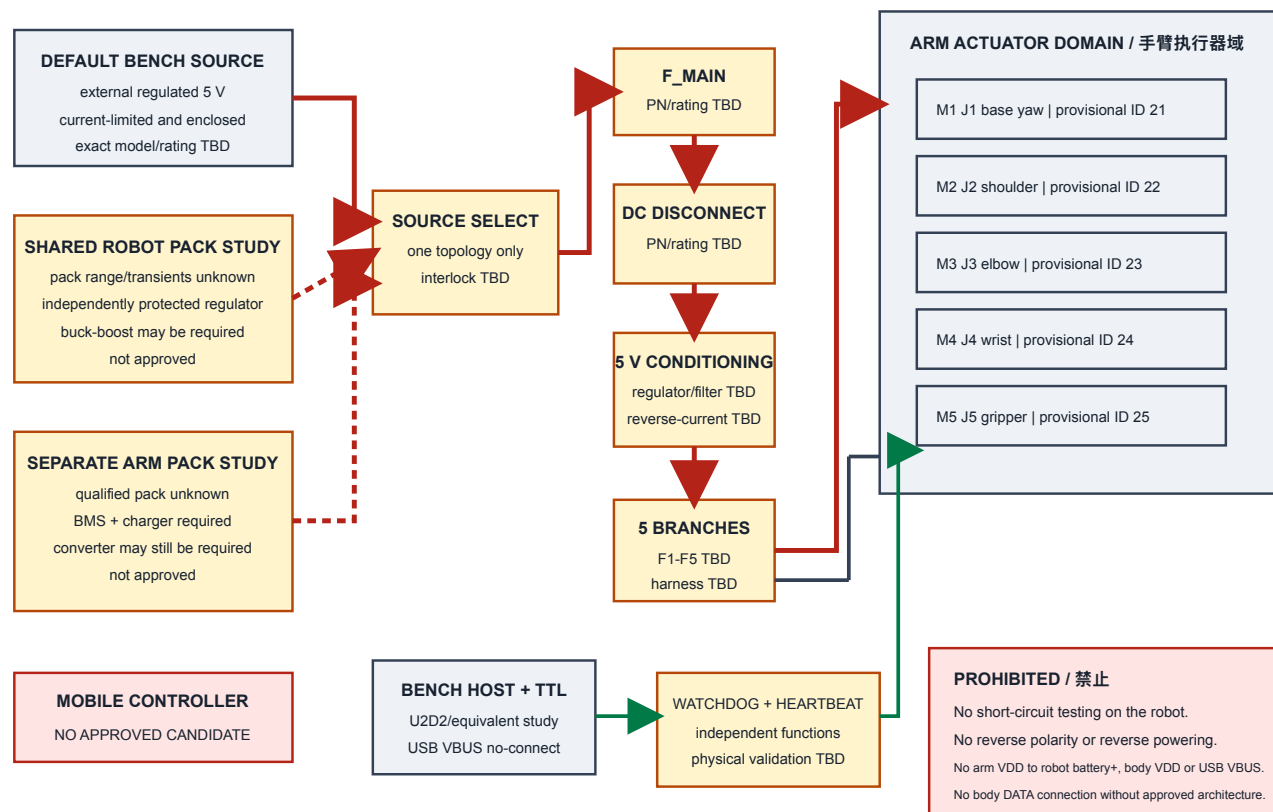


Figure 3: Power and data architecture

7.1 Bench default

The default qualification source is an external regulated 5 V bench supply with current limiting, enclosure, calibrated measurement, main protection, manual DC-rated disconnect, and individually reviewed actuator branches. No exact source, fuse, connector, wire, switch, or branch-protection part is selected.

7.2 Shared-pack study

A shared robot-pack topology may be studied only through an independently protected regulator. The robot pack's minimum, maximum, transient, load-step, and regenerative behavior are unknown. Therefore a buck-boost converter may be required; a buck-only converter must not be assumed from nominal voltage.

The retained Pololu D24V60F5/D24V90F5 mechanical files are vendor references, not qualification of voltage range, current, thermal performance, dynamics, mounting, or reverse-current behavior for this robot.

7.3 Separate-pack study

A separate arm pack remains conditional on exact chemistry, cell count, capacity, fault current, BMS, charger, enclosure, connector, transport controls, runtime, thermal behavior, regulator, and regenerative protection. No pack is qualified.

7.4 Prohibited connections and actions

- Never intentionally short the source or perform short-circuit tests on the robot.
- Never reverse polarity.
- Never connect arm VDD to robot battery positive, body VDD, or USB VBUS.
- Never permit the arm source or USB interface to reverse-power the robot.
- Never connect arm DATA to the body actuator bus without an approved architecture.
- Never infer power delivery from U2D2 enumeration.

There is no approved mobile controller. U2D2 or equivalent is only a bench communication study.

8. Safety and control gates

The current safety code is a deterministic simulation/HIL state model with no device I/O. It models:

- a 100 ms command deadline;
- a separate 100 ms software-heartbeat deadline;
- monotonic sequence and timestamp checks;
- replay detection;
- latched E-stop, overcurrent, and undervoltage faults;
- fail-closed motion authority.

Repeated read or heartbeat traffic cannot refresh stale command age. A bus watchdog requests a stop but does not guarantee torque-off. Physical E-stop, power isolation, contactor/disconnect behavior, fault detection, and real actuator stop time are unqualified.

The arm model uses a 15-action, 64-observation v1-C contract. No approved mobile controller implements the physical mapping. Hardware software must preserve deterministic J1-M1 through J5-M5 mapping, actual unique IDs, joint-sign and zero calibration, slew/current/temperature limits, watchdog, heartbeat, and explicit arm/disarm/reset state.

9. Staged hardware acceptance

The complete bilingual command procedure is `HARDWARE-SOP.md`; the planned cell-level matrix is `TEST-MATRIX.csv`.

Stage	Scope	Current state
S0	configuration and evidence audit	NOT RUN
S1	unpowered mechanical/ load-path inspection	NOT RUN

Stage	Scope	Current state
S2	isolated power-domain qualification	NOT RUN
S3	torque-disabled communications and safety logic	NOT RUN
S4	fixed-root single-axis dynamic motion	BLOCKED
S5	fixed-root coordinated motion and payload ladder	BLOCKED
S6	restrained-system disturbance/fault test and mobile decision	BLOCKED

Payload levels are P0, P5, P10, P20, P30, P40, and P50 in grams. P0-P10 are commissioning points. P20 is the stationary target. P30-P50 are optional engineering overload points, not ratings; they require separate authorization and may be omitted when unsafe.

Advancement requires traceable measurements and a reviewed pass at the prior stage and payload. No matrix cell has passed. v1-B outcomes and current simulation runs cannot fill physical evidence cells.

10. Release gates

Fabrication, powered bench motion, or mobile integration remains blocked until the applicable owner closes all of the following:

1. Physical robot inventory and exact actuator SKUs.
2. Official mating geometry, actual measurements, fit coupons, and controlled drawings.
3. Complete shaft, bearing, horn, spacer, retainer, fastener, bracket, and link definitions with structural calculations and proof loads.
4. Independent wrist and jaw transmission release.
5. Gear material, pressure angle, face width, backlash, strength, wear, retention, and guarding.
6. Measured pad dimensions, material, friction, compliance/contact stiffness, retention, and wear.
7. Closed mass ledger at or below 160 g, measured CoM, and measured or validated full inertia.
8. Qualified continuous torque-speed-thermal envelopes meeting the 3x gate.
9. Candidate A/B tracking, energy, thermal, backdrive, and disturbance comparison.
10. Qualified source/pack, BMS/charger where applicable, converter, reverse-current behavior, fuses, disconnect, connectors, wire, grounding, and transient/thermal evidence.
11. Physical mapping and calibration of unique bus IDs; provisional IDs do not count.
12. Independent watchdog, software heartbeat, physical E-stop, reset, and real stop-response validation.
13. Selected and reviewed mobile controller.
14. Sensor/estimator latency, attachment compliance, dynamic balance, collision, dropped-load, tip-over, and human-access validation.

11. Stop condition

This baseline is complete when it enables review without implying fabrication or test success. It intentionally stops before part selection, machining, wiring, powered motion, or mobile release. The next authoritative records are controlled drawings, selected-part evidence, completed S/P test records, and the leader-owned results files.

v1-C measured simulation results

NOT ALL EXPERIMENTS PASS. Software regression checks and the fixture task matrix pass; free-base manipulation/walking and physical release do not. Preserve those failures rather than relaxing acceptance.

Test evidence / 测试证据

298 passed, 11 warnings in 59.25s

Ruff and Python compilation pass. Standalone mypy remains blocked by resolving the workspace' s `microduck_local.contract` import; this is not claimed as a passed whole-project type check. Runtime policy-contract regression tests pass.

Mode	Case	Success / total	Failure
fixture	Obstacle move	20/20	PASS
fixture	Pick/place	20/20	PASS
fixture	Reach	20/20	PASS
fixture	Waypoint carry	20/20	PASS
free	Obstacle move	0/20	arm_environment, object_drop, phase_timeout
free	Pick/place	0/20	arm_environment, object_drop, phase_timeout
free	Reach	8/20	timeout
free	Waypoint carry	0/20	arm_environment, object_drop, phase_timeout
free	Walk (stowed)	0/20	timeout
free	Walk + carry	0/20	object_drop, robot_self_collision

Candidates A/B, seeds 0 - 9, 10 g object. Fixture results are **not** free-base qualification. The table top is 145 mm; v1-B low-table results are not directly comparable.

BC → DAgger → residual PPO

8 teacher episodes (seeds 0 - 7); 4 DAgger episodes (seeds 20 - 23). 20 BC epochs, 20 merged-data BC epochs, 2,048 PPO timesteps (seed 40). Held-out seeds: 101 - 105. Case: reach, fixture, candidate A. Five arm outputs with an explicit fifteen-action adapter; not stock 14-action ONNX.

Policy	Held-out success	Failures
bc	0/5	{ "arm_environment" : 5}
dagger	4/5	{ "timeout" : 1}
ppo	5/5	{}
teacher	5/5	{}

Final metrics (raw, not smoothed):

```
{  
  "bc": {  
    "epoch": 20,  
    "raw_train_arm_mse": 0.0018554778071120381,  
  },  
}
```

```

    "raw_validation_arm_mse": 0.005228061694651842
  },
  "dagger_merged_bc": {
    "epoch": 20,
    "raw_train_arm_mse": 0.0008629434159956872,
    "raw_validation_arm_mse": 0.0036217200104147196
  },
  "ppo": {
    "approx_kl": 9.514857083559036e-06,
    "clip_fraction": 0.0,
    "entropy_loss": -7.093945741653442,
    "phase": "training_end",
    "raw_loss": 0.05093790963292122,
    "raw_reward_sum": -1.2966566238901578,
    "timesteps": 2048,
    "value_loss": 0.104794941842556
  }
}

```

PPO retains the IK teacher with a maximum 0.02 rad arm residual. Matching the teacher's 5/5 is not evidence that PPO is better than the teacher. BC loss reduction does not imply closed-loop success. Dagger still has a failure.

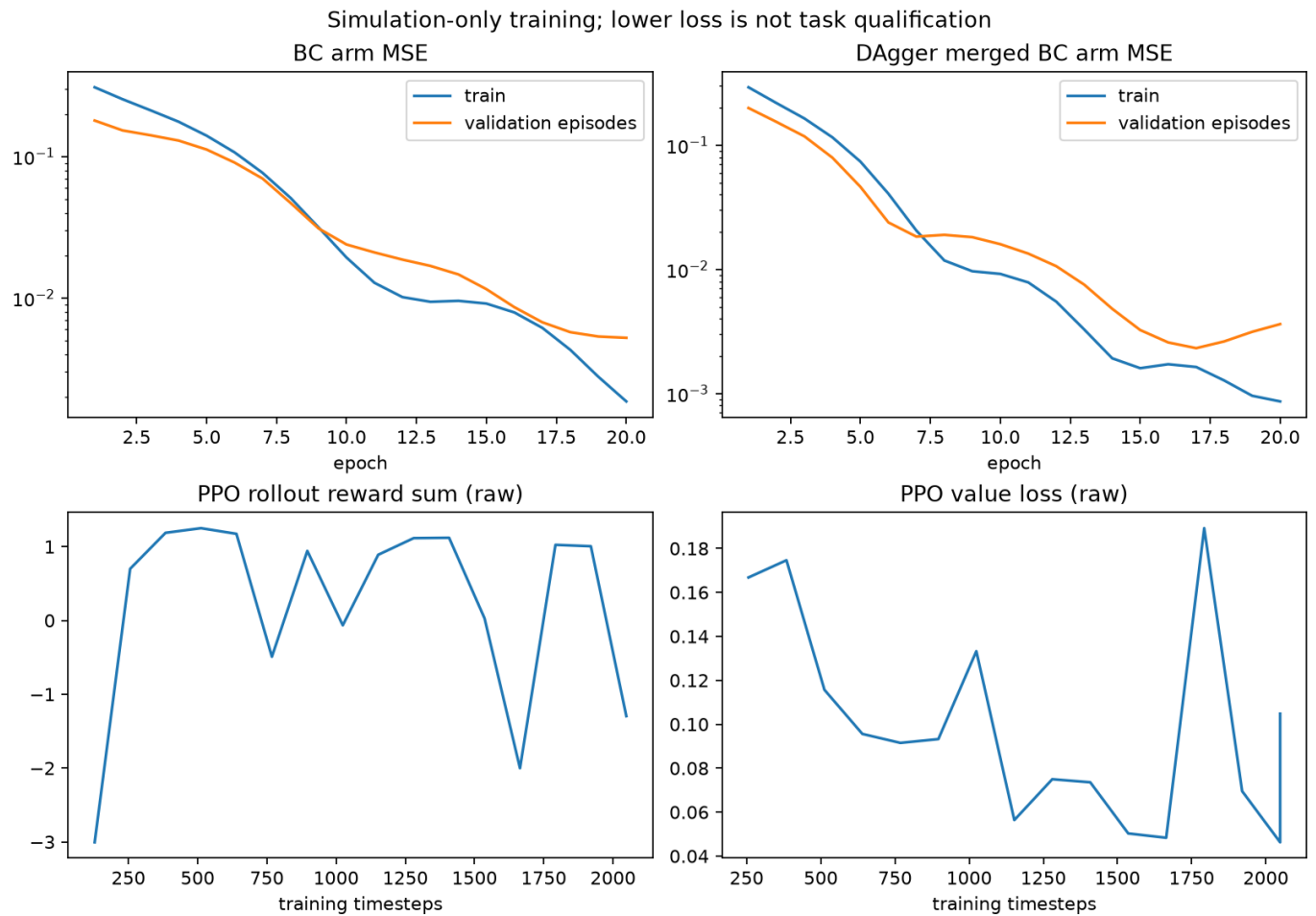


Figure 4: Raw learning curves / 原始训练曲线

Workspace / 工作空间

10000 sampled target positions; 3505 IK-reachable; 3255 reachable and collision-free. Balance-qualified workspace: **NOT ESTABLISHED**. Full mass/CoM/inertia tables are analytical model estimates, not measured CAD release. Joint torque CSV contains 5,000 screening rows; qualified continuous torque is unknown. The 3× torque release gate remains closed.

Payload screening / 负载筛查

Candidate A, fixture, seed 0 only: these are not payload ratings or reliability estimates.

Payload g	Case	Task success	Failure
5	Reach	True	PASS
5	Pick/place	True	PASS
5	Obstacle move	True	PASS
5	Waypoint carry	True	PASS
10	Reach	True	PASS
10	Pick/place	True	PASS
10	Obstacle move	True	PASS
10	Waypoint carry	True	PASS
20	Reach	True	PASS
20	Pick/place	False	object_drop
20	Obstacle move	False	object_drop
20	Waypoint carry	False	object_drop
30	Reach	True	PASS
30	Pick/place	False	object_drop
30	Obstacle move	False	object_drop
30	Waypoint carry	False	object_drop
50	Reach	True	PASS
50	Pick/place	False	robot_self_collision
50	Obstacle move	False	robot_self_collision
50	Waypoint carry	False	robot_self_collision

Videos / 视频

20 full IK/FSM episode videos, plus 4 actual ONNX policy videos. All are simulation recordings; successful and failed episodes are retained. Each video has full ffmpeg decode verification, ffprobe metadata and SHA-256. See [artifacts/microduck-arm-v1c/videos-final/video-manifest.json](#) and [artifacts/microduck-arm-v1c/videos-learned/video-manifest.json](#).

Release / 放行

Modeled cartridge: 166 g; target: 160 g — **FAIL**. Supervisor tests check emulated inputs, not real shorts/reversed wiring or device safety. Hardware release: **FALSE**. Fabrication, battery, continuous torque, thermal, sensor/estimator, mobile control and physical fault qualification remain blocked.

Final evidence directories: [evaluation-final](#), [engineering-final](#), [videos-final](#), [videos-learned](#), [learning/attempt-2](#) under [artifacts/microduck-arm-v1c/](#). Earlier root-level learning outputs, [engineering](#), and [videos-attempt-1-interrupted](#) are superseded diagnostics, not current release evidence.